

## Telecom electronics and infrastructure development

According to Tata Communications' Annual Report 2013-2014, in 2013, mobile devices accounted for 16 per cent and telecom infrastructure for nine per cent of the global telecom market.

The problem of inadequate infrastructure support in India led to the launch of shared mobile infrastructure scheme by Universal Service Obligation Fund (USOF) to provide subsidy support for setting up and managing 7353 infrastructure sites or towers spread across 27 states in the country where there were no existing fixed wireless or mobile network coverage.

When the scheme was closed at November 2013 end, 7317 towers had been set up. This infrastructure is now shared by three service providers with 16,254 base transceiver stations (BTS) commissioned for provisioning of mobile services.

**Test and measurement systems to support.** With an increase in the number of advanced communication networks and introduction of customised standards, need for advanced test and measurement (T&M) equipment required for research and development (R&D) labs, mobile devices, base stations and mobile backhaul, etc also increases. This includes spectrum and signal analysers, radio frequency and microwave communication testers, multiple input multiple output (MIMO) testers, oscilloscopes, signal generators and so on.

Imaging systems used for inspecting faults in electronic sub-systems of the telecom system is another field in testing that would grow with 3G, 4G and its successors.

### Making it in India

Reports from Directorate General of Foreign Trade (DGFT) for 2013-2014 show that the export and import of telecom electronic equipment in India (including mobile phones and their parts, cables, base stations, transmission and reception apparatus, data conversion and regeneration mechanism and switching systems) touched

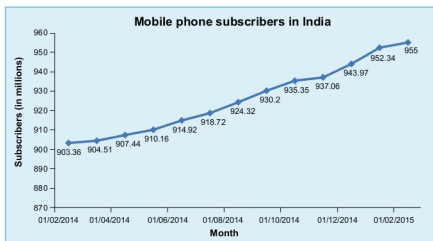


Fig. 1: Chart showing the growth of mobile subscribers that spurs mobile phone sales

₹ 195.79 billion and ₹ 685.09 billion, respectively. This shows the vast opportunities open to electronics components and devices manufacturers, and infrastructure providers in the country.

**The first step.** The Indian mobile phone brands Micromax, XOLO (Lava International), Karbonn, Spice and Celkon have been manufacturing their handsets in China and importing these to India.

According to reports from Press Trust of India, Micromax has started manufacturing mobile phones at its Rudraprayag plant in Uttarakhand, where they have already been manufacturing LED TVs and tablets. This is a giant leap towards the indigenous manufacturing movement started by the government of India.

**Going ahead.** Big mobile phone vendors, in the near future, are expected to start making their devices in India. This move towards local manufacturing can be seen from Samsung's announcement that the firm will start manufacturing Samsung S6 and S6 Edge smartphones in their two Indian manufacturing plants by second quarter of 2015.

Chinese manufacturer Xiaomi also has plans to start manufacturing in India within the next two years, as mentioned by the company's vice president, Hugo Barra, to Bloomberg.

The new in line is vivo Mobile that recently entered Indian market through a partnership with Viacom

18's integrated network solutions (INS). According to a report by Business Line, director of vivo Mobile India, Jacky Liao, mentioned that the company is keen on manufacturing their phones in India and will look at it in the next three years depending on its growth.

### Barriers: Infrastructure and competitors

While we find companies moving towards establishing their manufacturing units in India, handset giant Nokia Oyj shut down its plant in Chennai last year. Soon after this move, electronics contract manufacturer Foxconn Technology Group also suspended its operation in the country.

This could have a negative impact on the local manufacturing campaign. What we find here is a missing ecosystem that provides mobile phone components such as processors, displays, batteries and other parts.

Indian mobile industry is also concerned about the preference being given to Vietnam as a manufacturing hub due to its investor-friendly ecosystem, better incentives and other benefits from their government.

Manufacture of handsets in India, apart from creating employment opportunities, would also lead to cheaper, good quality phones. But in order for this to take off, the government needs to facilitate a component manufacturing ecosystem here.